FACULTY OF COMPUTING & INFORMATION TECHNOLOGY
KING ABDULAZIZ UNIVERSITY

FCIT
KAU

كلية الحاسبات وتقنية المعلومات
جامعة الملك عبدالعزيز

CPIT280 Project

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Problem definition :

Managing employee leaves, contracts, and attendance is a big challenge for the HR department. The current system relies heavily on manual work, which causes several issues. Manual processes lead to mistakes and delays in tracking leaves. Additionally, contract expirations may go unnoticed, resulting in operational and legal problems. Payroll data is often inaccurate due to human mistakes, and there are no automatic alerts for contract renewals, expirations, or leave approvals. Finally, finding data is slow because records are stored in Excel without a clear system. Moreover, updating data takes a lot of time, and sometimes employees forget to make the necessary changes ,This leads to inefficiency, more mistakes, and delays in HR tasks.

Interview Outline:

Field	**Details**
Interviewee	HR \ Haneen Almansouri
Company	Al-Qafila & Al-Yanabee Industrial Advertising Boards
Location	Jeddah, Second Industrial City
Industry	Manufacturing advertising boards for major companies such as Al-Tazaj, Al-Ahli Bank, KFC, Hardee's, Noon, Panda, and more, including street signage.
Interviewer	Tarfaa Alakour, Ghaida Hadi, Amal Al Juhani , Mawadda alsubhi
Medium	Online on teams
Start Time	5:00 PM
End Time	5:28 PM

Objectives:

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Objective Number	Description
1	Understand HR challenges in managing employee leave, contracts, and attendance.
2	Identify pain points in employee record tracking and update mechanisms
3	Gather insights on desired system features for automation.

Agenda:

Agenda Item	Duration	Description
Introduction	2 min	Explanation of the project and purpose of the interview.
Background on Project	3 min	Overview of the Leave Tracker System and its benefits for HR operations.
Overview of Interview	2 min	Discussion on topics to be covered and expected outcomes.
Interview Topics & Questions	15 min	HR daily tasks, challenges in employee data management, attendance and leave tracking difficulties, contract management issues, desired automation features.
Summary of Major Points	3 min	Recap of key insights and challenges discussed.
Questions from Interviewee	2 min	Address any queries or additional inputs from the interviewee.
Closing	1 min	Thanking the interviewee and explaining the next steps.

Interview Questions & Responses

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Question	Response
What are the main daily tasks handled by the HR department?	The HR department takes care of employees both inside and outside the organization, including recruitment, medical insurance, interviews, attendance tracking, and government affairs management.
How are employee data managed? Is there a clear system in place?	Employee data is stored in an Excel database. Previously, there was no unified system, and efforts were made to manually consolidate the data.
Do you rely on electronic systems, or is most of the work manual?	75% of the work is electronic, with limited manual processes for tasks such as leave and permission requests.
What information is required when hiring a Saudi vs. a foreign employee?	**Saudi:** CV, national ID, certificates, previous experience. **Foreign:** Residency details, sponsorship, visas.
Do you face difficulties retrieving employee data?	Retrieving data is not a problem, but keeping it updated and tracking contract expiration dates is challenging.
Are there any tasks that take too much time?	Calculating leave entitlements is time-consuming as the current system is paper-based and inefficient.
Do errors frequently occur in your work?	Yes, errors occur due to reliance on memory and paper-based systems, especially in leave calculations.

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Is it difficult to track attendance, leave, or salaries?	Yes, tracking depends on a single employee. If they are absent, confusion arises. There was an attempt to integrate attendance tracking into a system, but it was not completed.
Are there reports or processes you wish were automated?	Yes, leave, attendance tracking, and employee data updates should be automated.
If a leave management system is built, what features should it have?	It should include notifications for contract expirations and leave, along with automatic employee data updates.
If a contract management system is built, what features should it have?	A comprehensive database with employee details, automatic contract updates, and advance notifications for expiring contracts and residency permits.
Do you face difficulties in tracking attendance?	Yes, the system depends on a single employee. If they leave, confusion arises. There was an attempt to integrate attendance tracking with biometric devices, but it was unsuccessful.
Is Excel your primary tool for managing employee data?	Yes, Excel is widely used for employee and contract data, but there is a need for more advanced solutions.
What tasks take too long and need improvement?	Calculating leave entitlements and tracking contract renewals need improvement as they are manually handled and time-consuming.
If a new electronic system is introduced, what features should it include?	Automatic data updates, contract and leave notifications, a unified employee database, and quick data retrieval.
What challenges do you face in updating employee records?	The absence of an automatic update system makes it difficult to track expired contracts and pending entitlements.

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Is there a need for an internal notification system to alert employees about entitlements and leave?	Yes, an internal notification system is essential for alerting employees and managers about leave, contracts, and entitlements.
Is there a difference in handling data for Saudi and foreign employees?	Yes, foreign employees require additional procedures, such as sponsorship transfers and residency renewals.
Is there an intention to adopt a more advanced system instead of Excel?	Yes, there is a desire to transition to a more advanced system, especially for contract and leave management.
Do you face difficulties retrieving employee information when needed?	Yes, as searching for information in Excel is manual, and updates are not always consistent.
Do you need automated reports to track employee performance?	Yes, performance reports are very useful, particularly for attendance, leave, and entitlements.
What challenges do you face in contract updates?	The lack of a contract expiration notification system makes it difficult to track expiring contracts in time.

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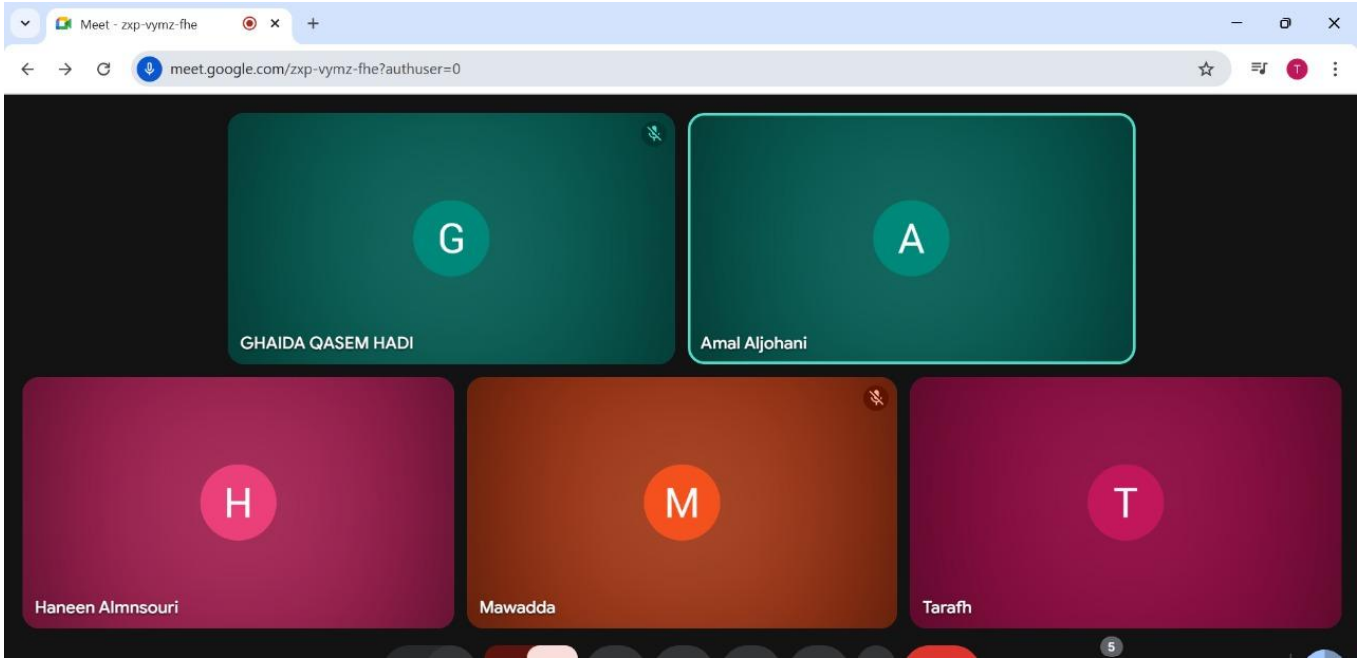

Persona :

Persona 1: Haneen – HR Manager (Traditional)

Name: Haneen Al-harbi
Age: 40
Occupation: HR Manager
Location: Jeddah, Saudi Arabia

Technology Comfort Level :

Haneen is not very comfortable with modern technology. She trusts paper records more than digital systems. She often prints everything out because she believes hard copies are safer.

Goals :

- Keep employee contracts and leave requests organized.
- Ensure no important information is lost.

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- Have physical proof for every decision.

Motivations :

- Feels more in control when she can see and hold paper files.
- Wants to avoid mistakes and maintain accuracy in records.
- Believes traditional methods reduce risks of “system failure.”

Frustrations :

- Finds digital systems confusing and time-consuming.
- Feels stressed when she has to depend only on software.
- Worries about losing data if the system crashes.

Daily Routine :

Morning: Checks pending requests in the system but prints them out.

Work hours (9 AM – 5 PM): Reviews paper copies of contracts and leave balances, signs approvals manually.

Afternoon: Updates files in folders, keeps physical records for security.

Evening: Leaves work satisfied only after all papers are in order.

Relevant Quote :

“I can only trust what I see on paper. A system can fail, but paper stays safe in my hands.”

Persona 2: Reem – HR Officer (Tech-Savvy)

Name: Reem Al-Harbi
Age: 30
Occupation: HR Officer

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Location: Jeddah, Saudi Arabia

Technology Comfort Level :

Reem is very comfortable with technology. She prefers using mobile apps, dashboards, and automated systems instead of paperwork.

Goals :

- Approve or reject leave requests instantly from her dashboard.
- Receive real-time contract expiration alerts.
- Save time by generating automatic reports.

Motivations :

- Loves efficiency and speed.
- Wants to reduce human errors by relying on automation.
- Feels proud when employees get faster responses and smoother service.

Frustrations :

- Gets annoyed if the system is slow or has bugs.
- Dislikes when data syncing is delayed.
- Frustrated if reports are not customizable.

Daily Routine :

Morning: Logs into the system on her laptop or tablet, checks dashboard alerts.

Work hours (9 AM – 5 PM): Approves requests, renews contracts, generates reports automatically.

Afternoon: Answers employee inquiries using data from the system.

Evening: Logs out after making sure all requests are processed digitally.

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Relevant Quote :

"With this system, everything is faster and smarter. I don't waste time on papers anymore."

Scenario :

Scenario: "A Smart Workday with Reem Al-Harbi"

1. Morning – Login and Overview
   Reem logs into the HR system from her tablet while sipping her morning coffee. The dashboard instantly loads, showing pending leave requests and upcoming contract expirations. She feels ready to take charge of the day digitally.

2. Approving Requests
   On her sleek interface, three leave requests appear. With one tap on "Approve," she finalizes them all — no paperwork, no delay.

3. Contract Alert
   A notification pops up: "Contract Expiring in 14 Days – Renew for Sarah M?"
   Reem reviews the details, clicks "Approve," and the system automatically generates the renewal contract for digital signing.

4. Auto Reports
   Later, she opens the "Reports" tab, selects the week's timeframe, and watches as colorful charts and summaries appear automatically. One click, and the report is emailed to management.

5. Employee Inquiry
   In the afternoon, an employee calls asking about their request. Reem checks the app on her phone — the data syncs instantly.

   "Your request is approved and logged," she replies confidently.

6. End of Day
   Before logging out, Reem glances at the system message: "All requests completed successfully." She smiles and thinks,

   "With this system, everything is faster and smarter."

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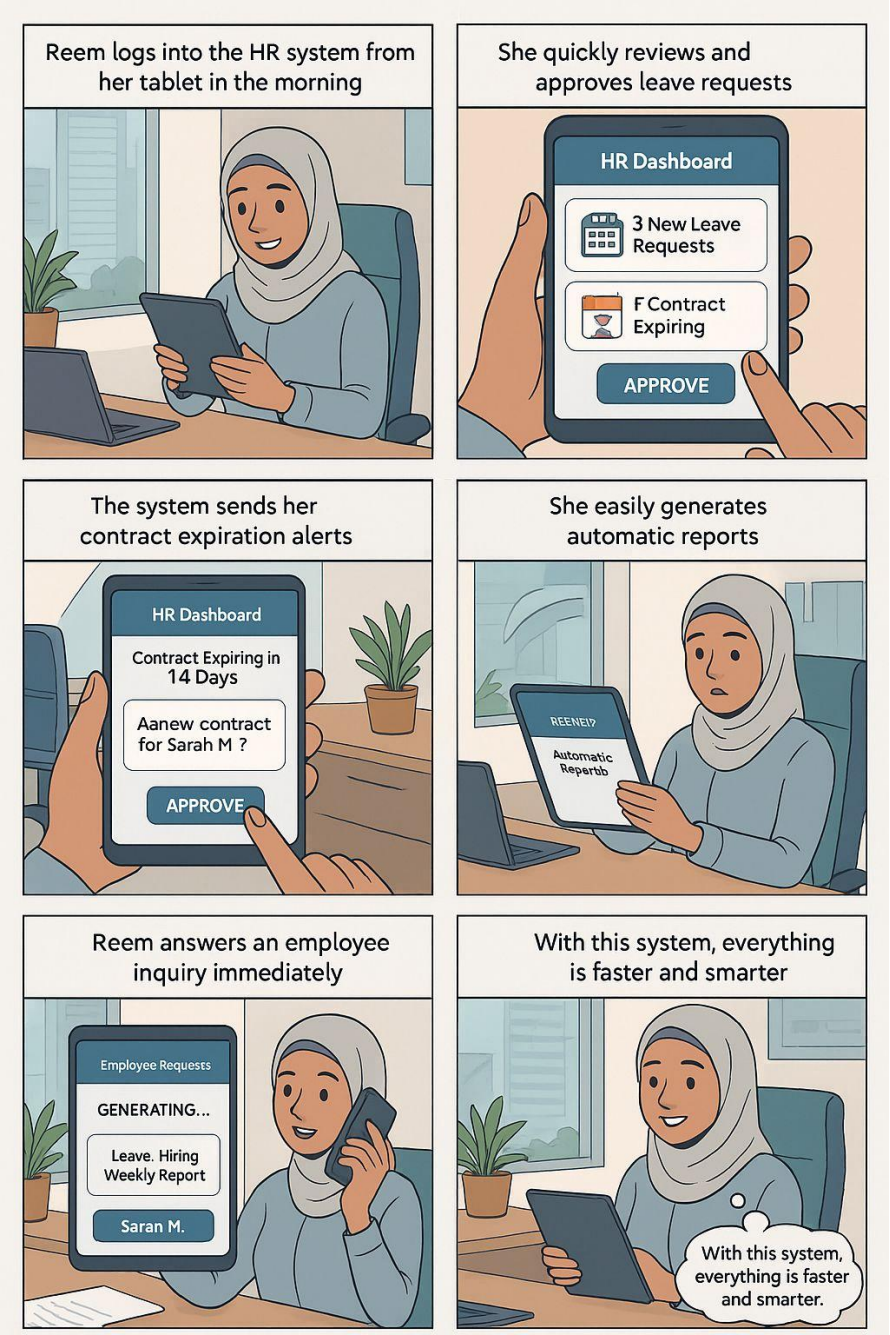
Storyboard:



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Functional Requirements:

Employee Information Management:

1. HR Staff shall add and update employee records (ID, name, department, joining date).
2. HR Staff shall search for and view full employee details by ID or name.
3. The System shall validate and require all mandatory employee data fields.

Leave Entitlement & Tracking:

1. The System shall automatically calculate and display available leave entitlements.
2. HR Staff shall record approved/rejected leave requests and the type of leave.
3. The System shall automatically deduct approved leave from the employee's balance and maintain a historical log.

Contract Management:

1. HR Staff shall enter contract details (start/end date) and attach the contract document.
2. The System shall display the remaining time until contract expiry.
3. The System shall alert HR Staff when a contract is 90 days or less from expiration.

Reporting:

1. HR Staff shall generate Employee Information, Contract Expiry, and Leave Reports.
2. HR Staff shall filter reports based on user-selected date ranges (e.g., contracts, leave).
3. The System shall export all generated reports into PDF and Excel/CSV formats.

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Non-Functional Requirements:

Performance & Technical:

1. The System shall be a web-based application accessible via standard browsers.

2. The System shall load primary screens and reports within 3 seconds.

3. The System shall use a Relational Database (RDBMS) and handle 20 concurrent users.

Security & Maintainability:

1. The System shall require unique username/password authentication for all users.

2. The System shall implement Role-Based Access Control (RBAC) for sensitive data.

3. The System shall log all critical actions (e.g., updates, adjustments) and provide daily automated backups

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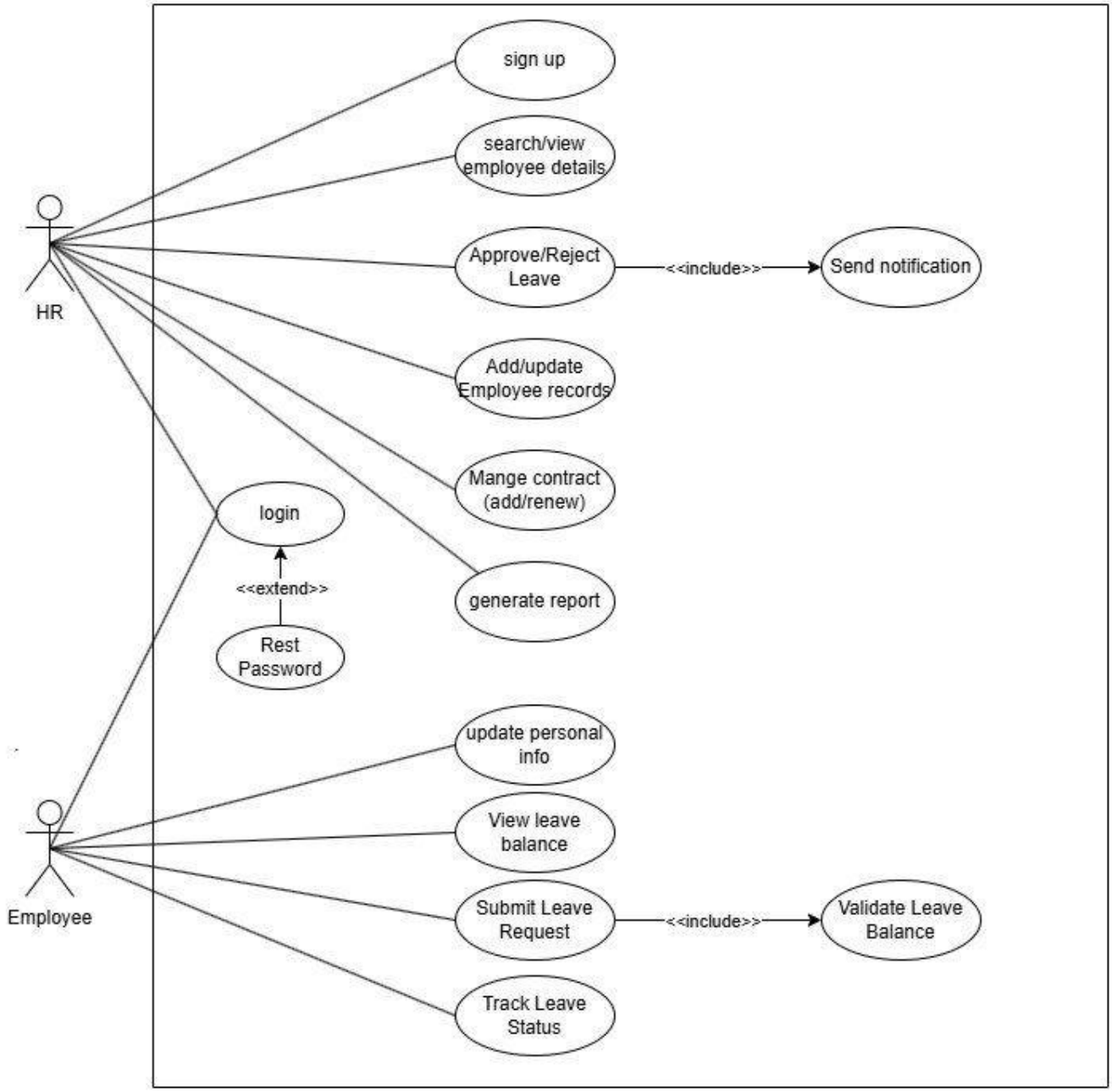
Use Case Diagram:



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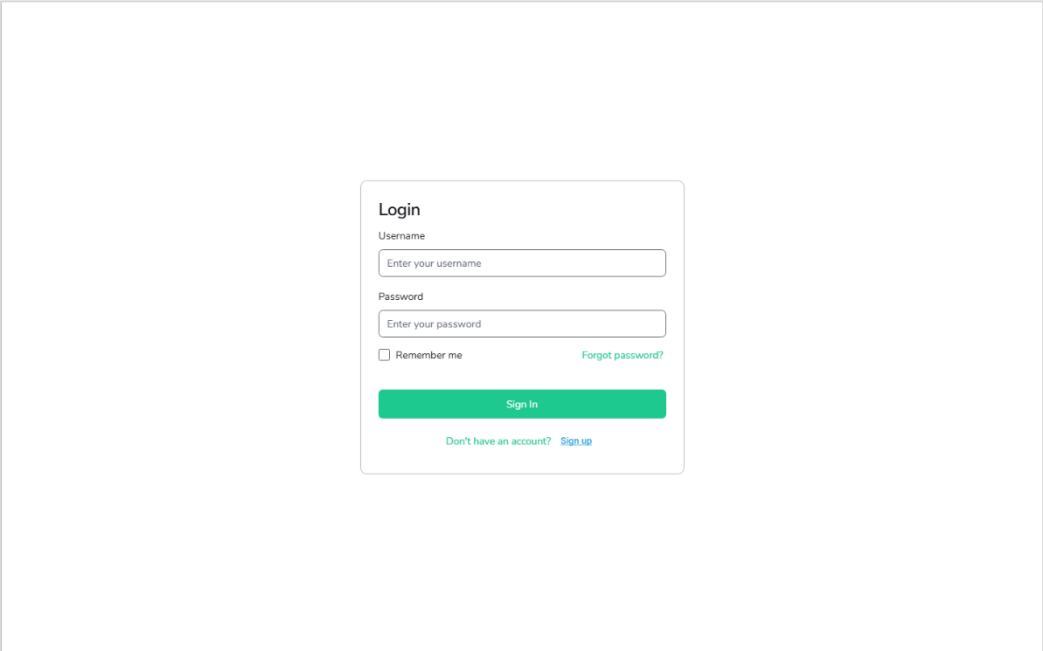
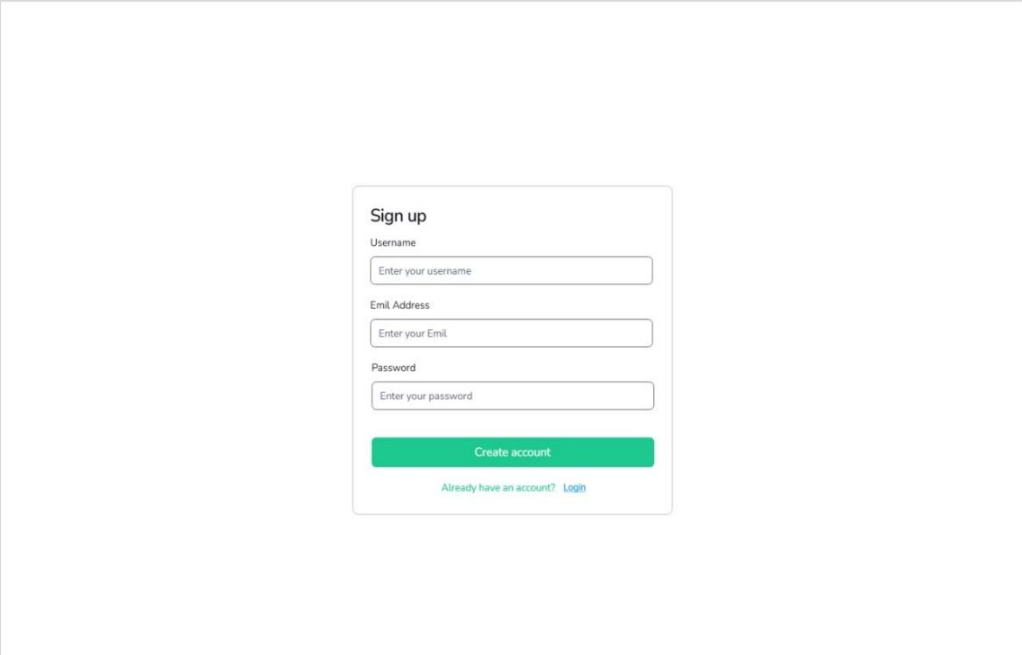
Wireframes:





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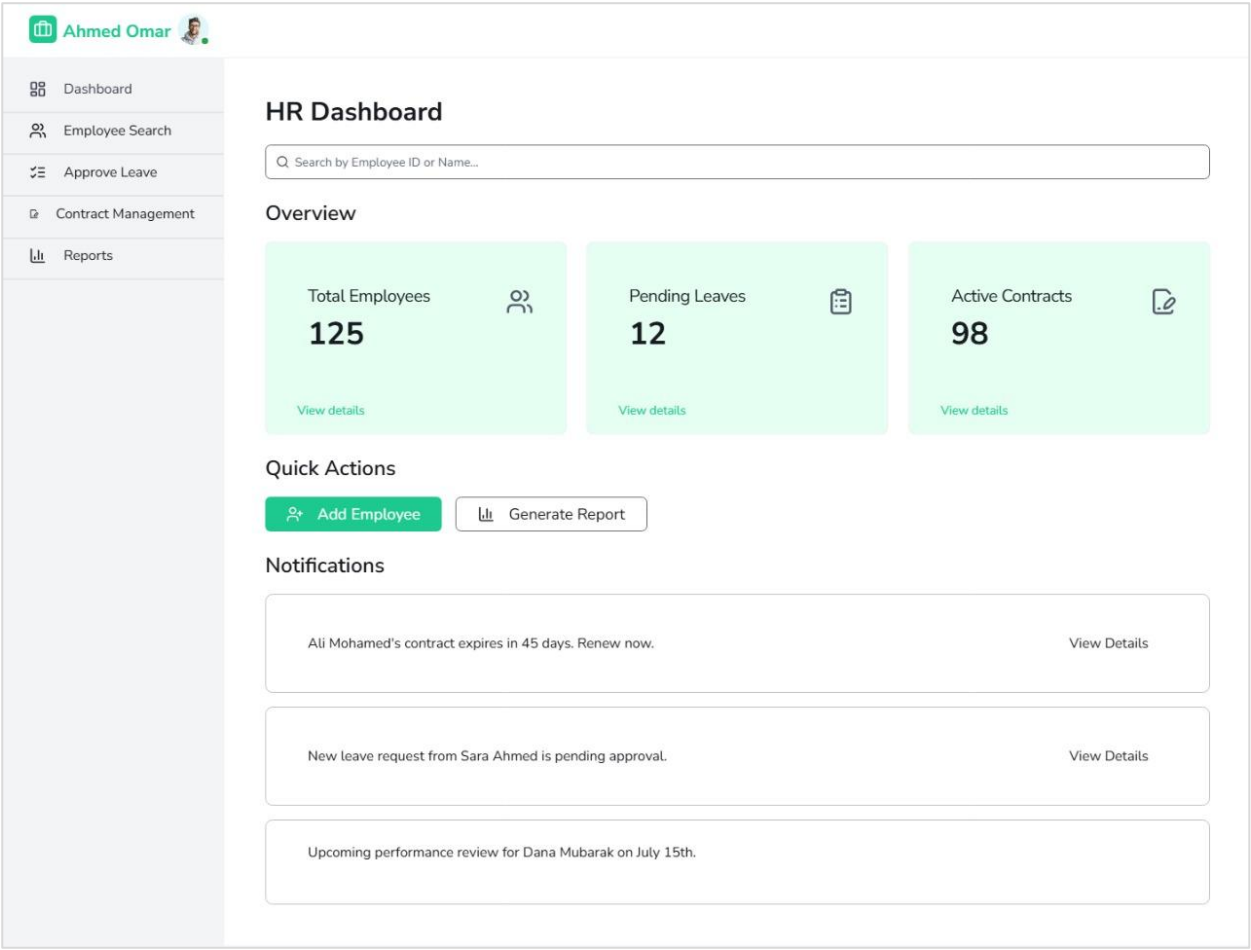

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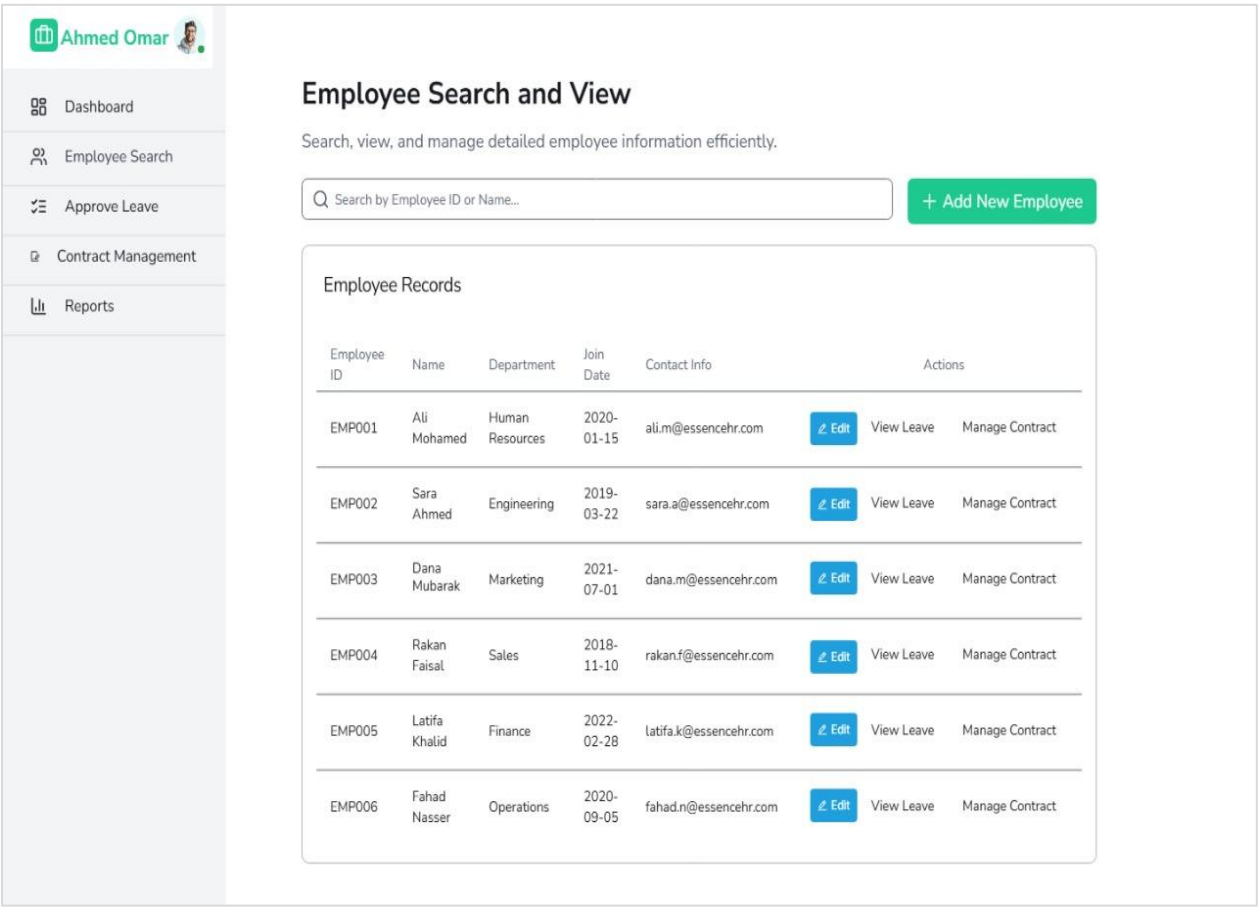

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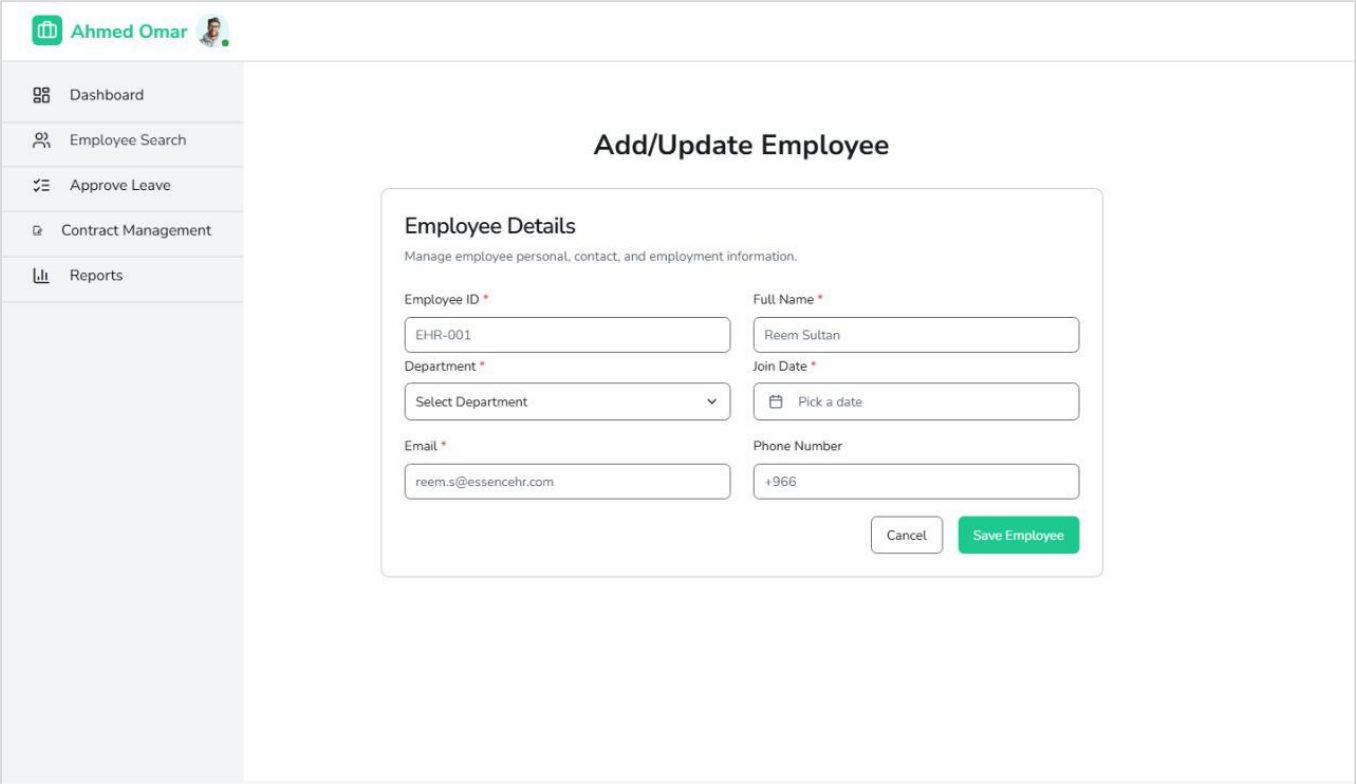

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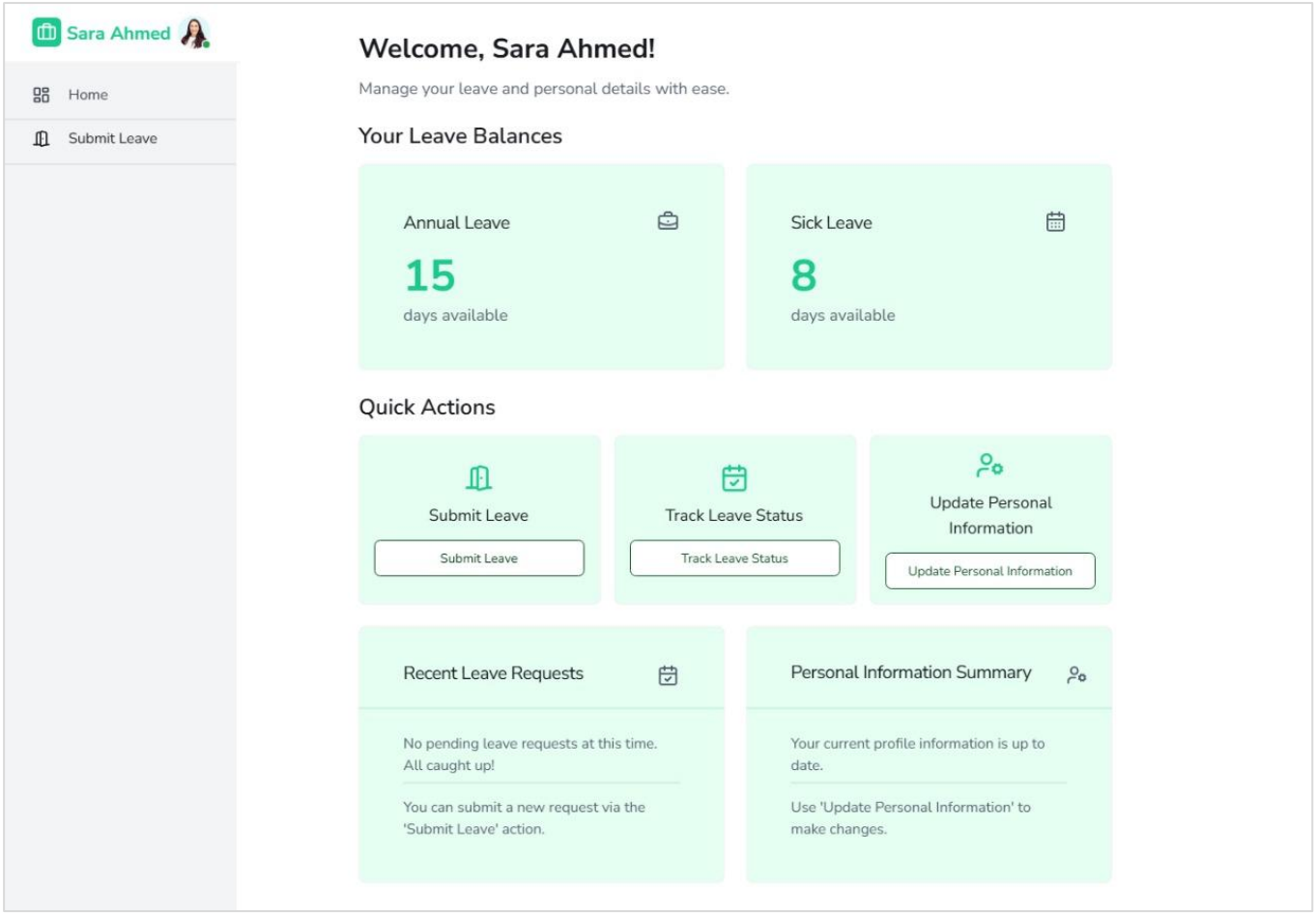

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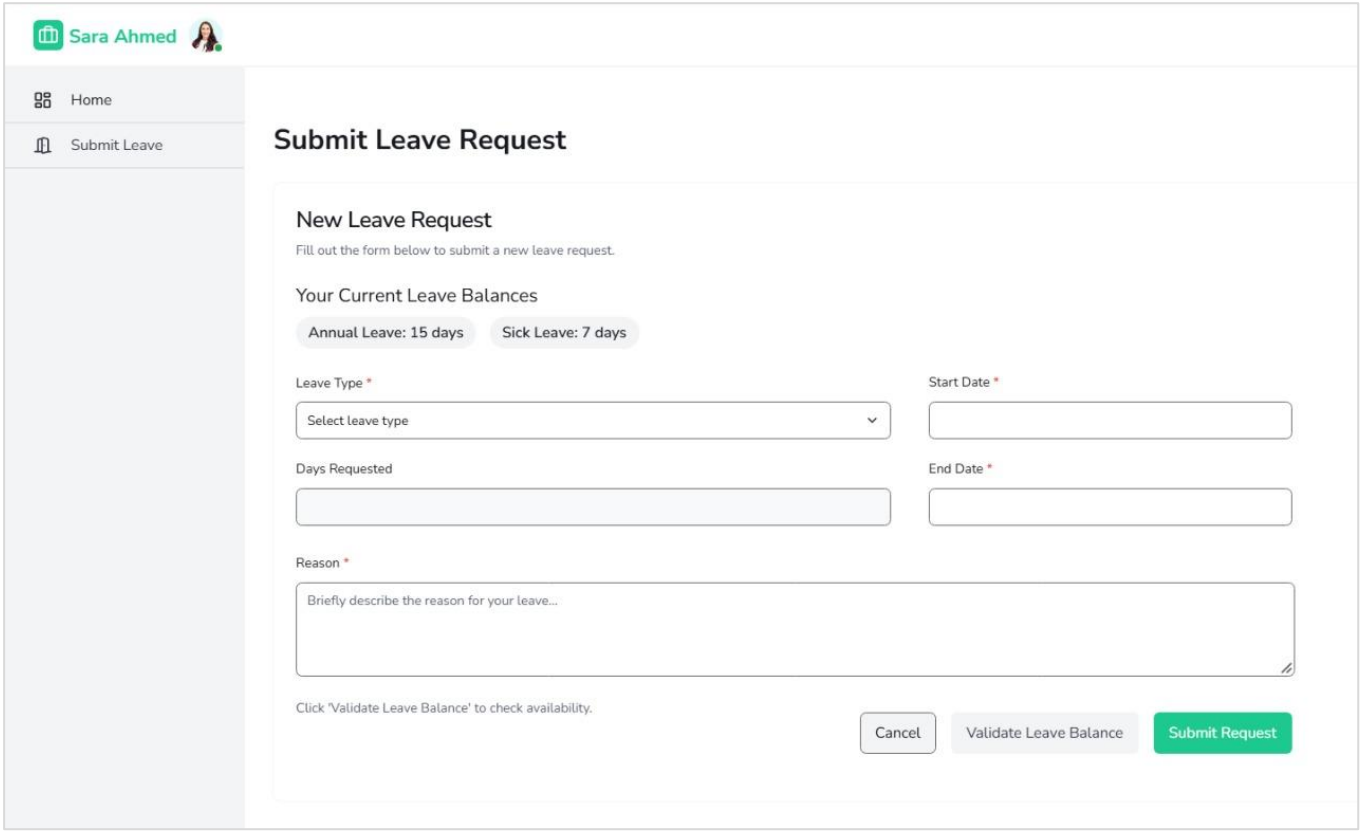

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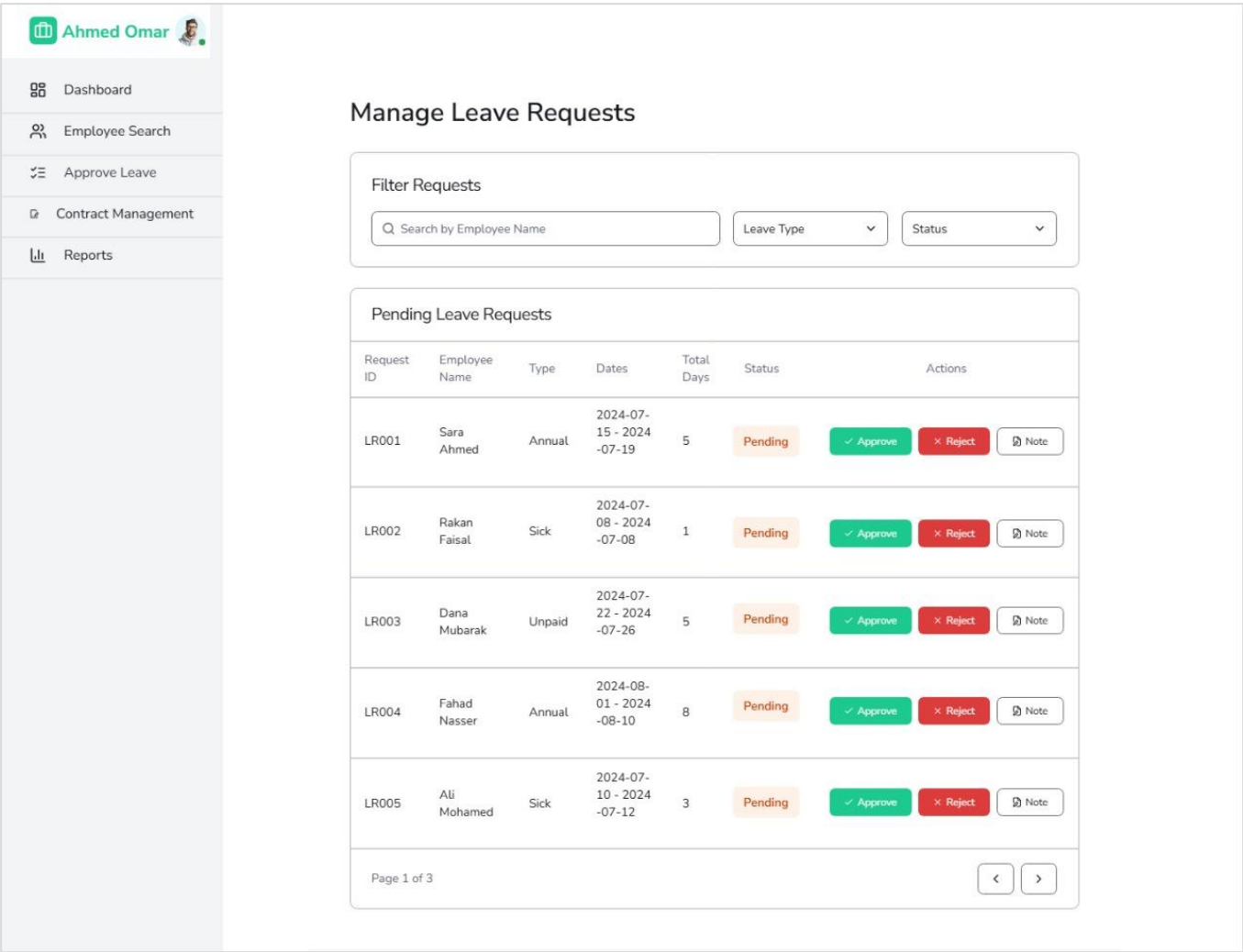

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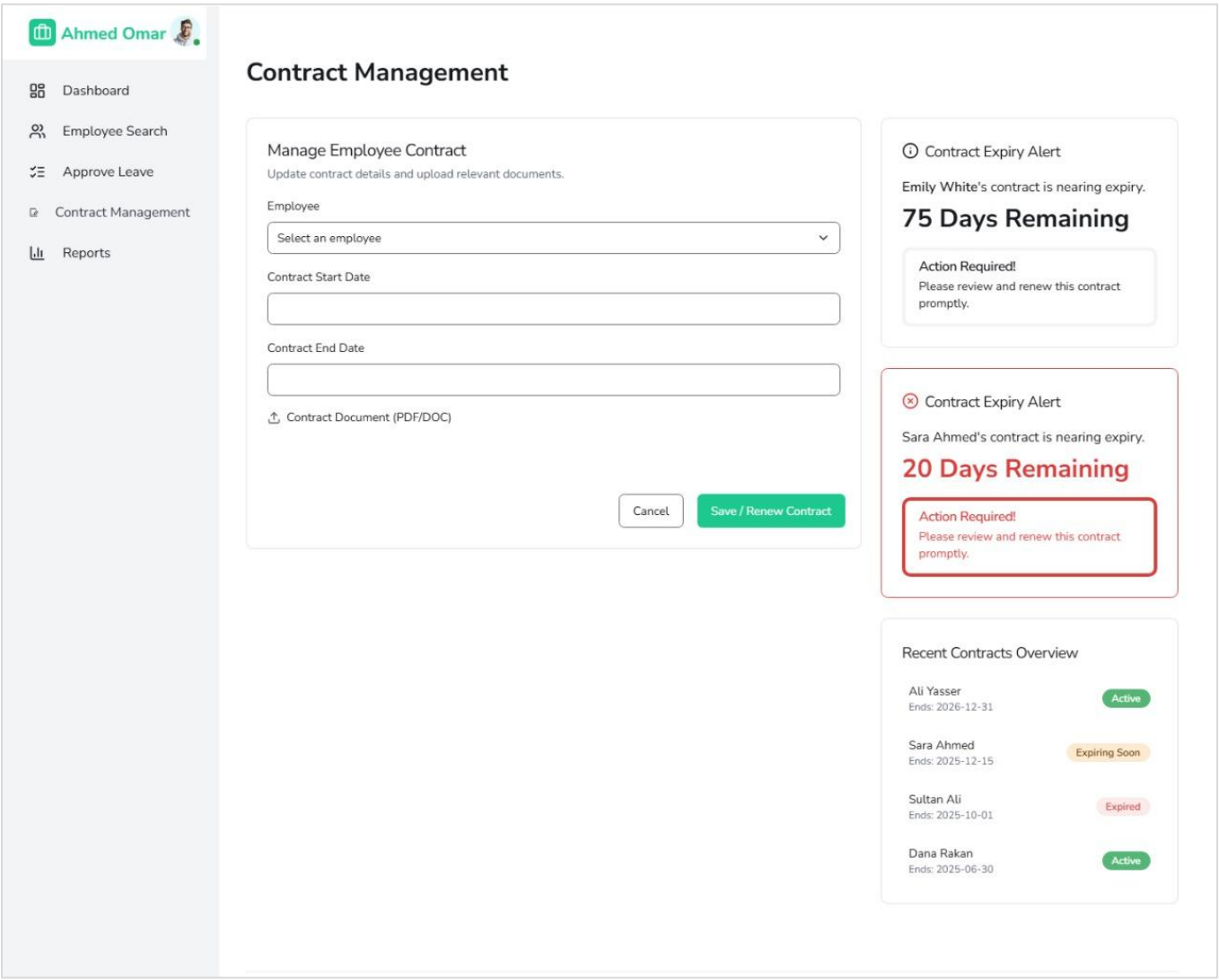

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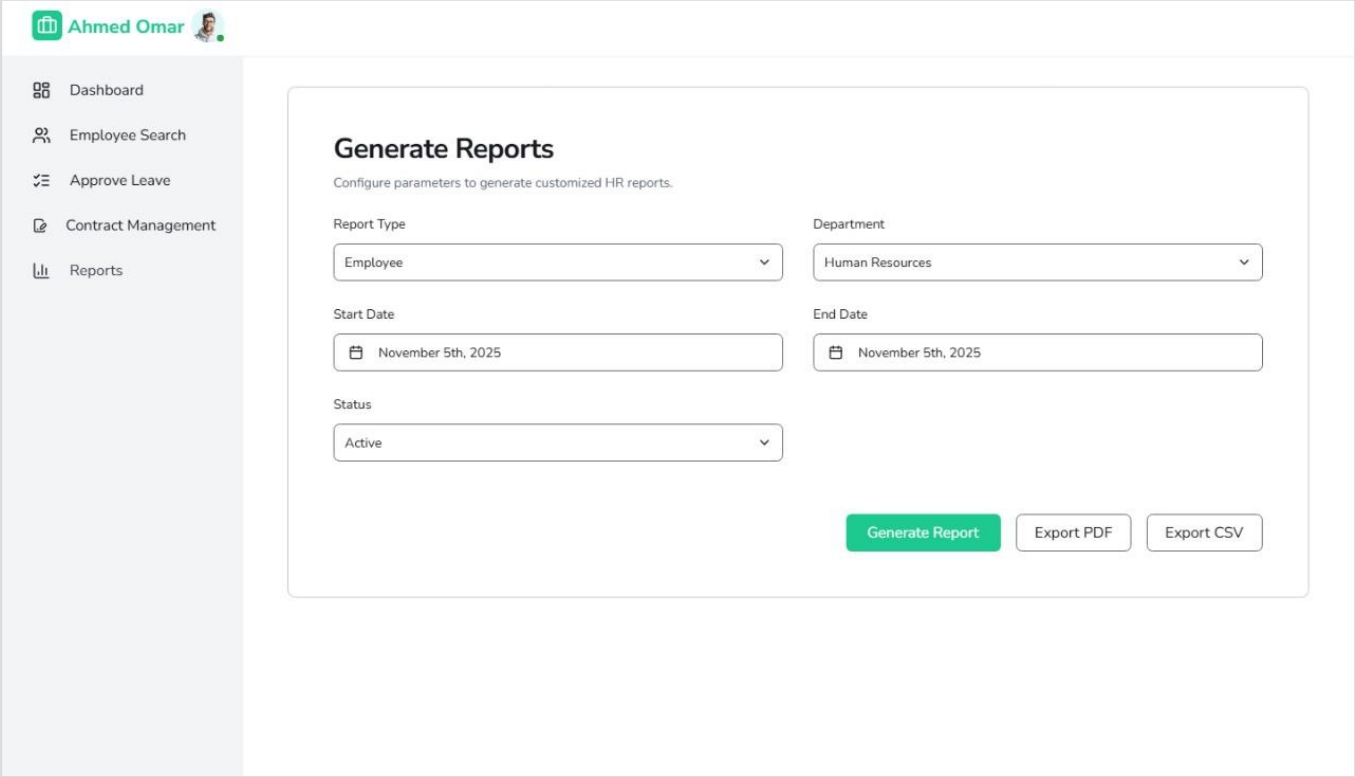

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Milestone 3: High-Fidelity Prototype

In Milestone 2, our team created the low-fidelity prototype using Visily, where we designed the initial Employee and HR flows for Milestone 3. However, after conducting internal testing and reviewing the design based on the 8 golden rules, we identified several issues. The Visily prototype didn't prevent user errors and lacked clear feedback, which we believed would negatively affect users during the testing stage.

To improve this, we exported the Visily design and rebuilt it in Loveable. The final high-fidelity prototype was redesigned to follow the 8 golden rules fully and to provide a more complete and interactive experience for our users.

Prototype Links:

1) Initial High-Fidelity Prototype – with issues (Visily):

Employee Flow:

https://app.visily.ai/projects/515d821e-76a5-47b8-87b9-b4df32f24555/boards/2332006/presenter?flowId=6216f1a3-3287-464a-b813-58f4ac33fc85

HR Flow:

https://app.visily.ai/projects/515d821e-76a5-47b8-87b9-b4df32f24555/boards/2332006/presenter?flowId=f63279e5-b73e-46e2-91bf-0520b486c736

2) Final High-Fidelity Prototype (Loveable):

https://hr-ease-system.lovable.app

Login Credentials for Testers (Loveable):

Testing Employee:

TESTEMPLOYEE@gmail.com

password123

Testing HR:

TESTHR@gmail.com

password123

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